

Rashtreeya Sikshana Samithi Trust

School of Computer Science and Engineering

RV UNIVERSITY

Bengaluru – 560059



Course Name: Computer Graphics and Virtual Reality
Program: B. Tech – CSE (H)
Course Code: CS4104
Semester: VII

Lab Instruction Manual

Academic Year: 2026–27

Vision

The School of Computer Science and Engineering at RV University envisions the Operating Systems course as a cornerstone of computing education, equipping students with a deep understanding of system architecture, resource management, and process coordination. This course fosters a strong foundation in theoretical principles and practical applications, enabling students to design, analyze, and optimize modern operating systems. With exposure to hands-on projects, research-led education, and industry linkages, the aspiration is to develop problem solvers and innovators who can create computing solutions that are efficient, secure, and scalable. Technological trends like virtualization, cloud computing, and cybersecurity are imparted in a way that prepares students to flourish in academia, industry, and entrepreneurship, thereby powering the advancement of technology and making a difference in society.

Mission

- * Fundamental Understanding- An OS itself is a rich set of concepts involving process management, memory management, file systems, and security under which students would have a considerable theoretical foundation.
- * Hands-On Learning- Students will have hands-on experiences with lab experiments, coding assignments, and project work involving actual systems.
- * Industry Relevance- Teaching concerning current industry practices offers students some understanding of topics in virtualization, cloud computing, and cybersecurity.
- * Research & Innovation- A research perspective fosters a contribution towards solving system-level problems and allows an exploration of potential emerging areas in OS development.
- * Performance Optimization- Students learn how to analyze and optimize the OS component for efficiency, scalability, and reliability.
- * Interdisciplinary Approach- Exploring the OS interplay with computer architecture, embedded systems, and networking helps students develop an understanding that is free from hardware-software divides.
- * Security & Reliability- Systems must be designed to be secure and to tolerate failures to protect against data corruption and system integrity.
- * Collaboration & Teamwork- Collaboration fosters innovation and teamwork skills in students through group projects, open-source contributions, and hackathons.
- * Ethical Computing- Awareness of the ethical considerations of system design including privacy, fair allocation of resources, and sustainable computing must be developed in students.
- * Career Readiness- A broad range of skills, knowledge, and experiences are provided through internships, collaboration with industry, and skill-build workshops to prepare students for career opportunities in the fields of software engineering, system administration, cybersecurity, and research.

Nomenclature for Program Educational Objectives (PEOs):

PEO1: Graduates will be able to demonstrate excellence in the field of Computer Science and Engineering through advanced research and entrepreneurship.

PEO2: Graduates will be able to achieve excellence in innovation through national and international academic and industry collaborations.

PEO3: Graduates will be able to actively engage in cutting-edge interdisciplinary and multidisciplinary research.

PEO4: Graduates will be able to function effectively as a talented pool of ethical, sustainable, self-driven, and life-long empathetic problem solvers.

Nomenclature for Program Outcomes (POs):

Engineering Graduates will be able to:

PO1: Engineering Knowledge. Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO2: Problem Analysis. Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO3: Design/Development of Solutions. Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for public health and safety, and cultural, societal, and environmental considerations.

PO4: Conduct Investigations of Complex Problems. Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO5: Modern Tool Usage. Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO6: The Engineer and Society. Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues, and the consequent responsibilities relevant to the professional engineering practice.

PO7: Environment and Sustainability. Understand the impact of professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO8: Individual and Teamwork. Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO9: Ethics. Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO10: Communication. Communicate effectively on complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO11: Project Management and Finance. Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO12: Life-Long Learning. Recognize the need for and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Nomenclature for Course Outcomes (COs):

Course Outcomes: After completing the course, the students will be able to:

CO1: Understand the fundamental concepts of computer graphics, graphics pipeline architecture, and graphics hardware systems.

CO2: Apply 2D and 3D graphics algorithms including clipping, projection, hidden surface removal, and shading techniques.

CO3: Develop real-time rendering applications using shader programming and modern graphics APIs.

CO4: Design and implement basic VR/AR interactive applications for real-world use cases.

Course Outcomes to Program Outcome Mapping:

Course Outcomes		PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
1.	Understand the fundamental concepts of computer graphics, graphics pipeline architecture, and graphics hardware systems.	3	2			2							
2.	Apply 2D and 3D graphics algorithms including clipping, projection, hidden surface removal, and shading techniques.	3	3	2	2	3							
3.	Develop real-time rendering applications using shader programming and modern graphics APIs.	2	3	3	3	3							2
4.	Design and implement basic VR/AR interactive applications for real-world use cases.	2	2	3	3	3							3

Note: High: 3, Medium: 2, Low: 1

SOFTWARE REQUIREMENTS

Operating System

- Windows 10/11 (64-bit)
- Ubuntu Linux 22.04 or above

Programming Languages

- C++
- C#
- GLSL

Software Tools

- Microsoft Visual Studio 2022
- Code::Blocks
- Visual Studio Code
- OpenGL
- FreeGLUT
- GLEW
- GLFW
- Unity Hub
- Unity 2022 LTS

HARDWARE REQUIREMENTS

- Intel Core i5 or above
- 8 GB RAM (Minimum)
- NVIDIA GTX/RTX or AMD Radeon Graphics (Recommended)
- 500 GB SSD
- 21-inch LED Monitor

Lab Marks Rubrics:

Table-1

Practical Execution rubrics (Max: 6 marks)					
Sl. No	Criteria	Measuring Methods	Excellent	Good	Poor
i.	Understanding of problem statement (2 Marks)	Observations	The student exhibits a thorough understanding of the requirements for the problem. (2 M)	The student has a sufficient understanding of the requirements for the problem. (1 M)	The student does not have a clear understanding of the requirements for the problem. (0 M)
i.	Execution (2 Marks)	Observations	The student demonstrates the execution of the program with optimized code and shows the output with appropriate validations and test cases handled. (2 M)	The student demonstrates the execution of the program without optimization of the code and shows the output. (1 M)	The student has not executed the program. (0 M)
i.	Results and Documentation (2Marks)	Observations	Documentation with appropriate comments and output cases are noted in data sheets. (2 M)	Documentation with few comments and few output cases are noted in data sheets. (1 M)	Documentation with no comments and no output cases are noted in data sheets. (0 M)
VIVA / Quiz Rubrics (Max: 4 marks)					
Sl. No	Criteria	Measuring Methods	Excellent (2 Marks)	Good (1 Mark)	Poor (0.5 Marks)

1.	Conceptual Understanding of Operating System (2 Marks)	Quiz	Thorough explanation of concepts	Adequate explanation with some gaps	Unable to explain tools/concepts clearly
2.	Application of Problem-Solving Techniques (2 Marks)	Quiz	Describes and justifies the selection of specific services	Identifies applicable methods with minor gaps in justification	Lacks clarity or incorrect application

Course Policy:

1. Materials Requirement:
 - a. All students must bring handwritten lab programs in datasheets for each lab session.
 - b. The datasheets should be properly maintained in an organized folder.
 - c. They are required to bring essential stationery, and a notebook for each lab session.
2. Submission Policy:
 - a. Datasheets will be evaluated and scored after each program execution.
 - b. Students must upload a scanned copy (softcopy) of the evaluated datasheet at the end of the lab session in Google Classroom.
 - c. The folder with all datasheets must be submitted at the end of the semester for final verification.
3. Academic Integrity Policy:
 - a. The use of mobile phones is not allowed during lab hours.
 - i. Also, they should not be connected to college WiFi.
 - b. Punctuality is important. Latecomers may not be admitted to the lab.
 - c. Any instance of academic malpractice discovered during assessments will result in a score of zero for that particular evaluation. Repeat occurrences will be escalated to the disciplinary committee for further action.
 - d. Consuming food or beverages inside the lab is strictly prohibited.
 - e. Any form of misconduct in the lab will result in disciplinary action and will be dealt with severely.

List of Laboratory Programs:

Sl. no	Programs
1.	Implement DDA and Bresenham line drawing algorithms
2.	Implement the Midpoint Circle Drawing Algorithm
3.	Implement 2D transformations (translation, rotation, scaling, reflection and shearing) using homogeneous coordinates
4.	Implement composite transformations using matrix representation
5.	Implement the Cohen–Sutherland line clipping algorithm
6.	Implement the Z-buffer algorithm for hidden surface removal
7.	Implement the Phong illumination model for a 3D object
8.	Develop a basic OpenGL program using vertex and fragment shaders
9.	Create a simple interactive 3D scene using Unity
10.	To implement the 3D viewing pipeline, including projection and viewport mapping for displaying 3D objects on a 2D screen

Laboratory Programs

EXPERIMENT – 1

IMPLEMENTATION OF DDA AND BRESENHAM LINE DRAWING ALGORITHMS

Aim

To implement the Digital Differential Analyzer (DDA) algorithm and Bresenham's Line Drawing Algorithm for generating straight lines on a raster display using OpenGL.

Course Outcome

CO1: Understand the principles of graphics systems, graphics primitives, and raster graphics algorithms.

Bloom's Taxonomy Level

L3 – Apply

Learning Objectives

After completing this experiment, the student will be able to:

- Understand raster scan graphics.
- Differentiate between DDA and Bresenham algorithms.
- Draw straight lines using incremental techniques.
- Compare floating-point and integer arithmetic algorithms.
- Analyze the computational efficiency of line drawing algorithms.

Output

Viva Questions

1. What is raster graphics?
2. Why are line drawing algorithms required?
3. Define scan conversion.
4. Explain DDA algorithm.
5. What are the drawbacks of DDA?
6. Why is Bresenham faster?
7. What is the decision parameter?
8. Why does Bresenham avoid floating-point arithmetic?
9. Which algorithm is more suitable for hardware implementation?
10. Compare DDA and Bresenham.
11. What happens when slope > 1 ?

12. What is aliasing?
13. Define pixel resolution.
14. Explain graphics primitives.
15. Give real-life applications of line drawing algorithms.

Assignment Questions

1. Modify DDA to draw dashed lines.
2. Implement dotted line drawing.
3. Draw thick lines using Bresenham.
4. Draw multiple connected line segments.
5. Compare execution time of both algorithms for 1,000 random lines.
6. Implement Bresenham for all eight octants.
7. Animate the line drawing process.
8. Draw a wireframe house using line drawing algorithms.

Mapping to Syllabus

This experiment addresses **Module 1** topics on graphics primitives and line drawing algorithms, specifically DDA and Bresenham line generation, as prescribed in the course outline.

EXPERIMENT – 2

IMPLEMENTATION OF MIDPOINT CIRCLE DRAWING ALGORITHM

Aim

To implement the Midpoint Circle Drawing Algorithm for generating a circle on a raster display using OpenGL by exploiting the eight-way symmetry of a circle.

Course Outcome

CO1: Understand and implement fundamental computer graphics algorithms for raster displays.

Bloom's Taxonomy Level

L3 – Apply

Learning Objectives

After completing this experiment, students will be able to:

- Understand the mathematical representation of circles.
- Apply the Midpoint Circle Drawing Algorithm.
- Utilize the eight-way symmetry property of circles.
- Compare circle generation techniques.
- Develop OpenGL programs for circle rendering.

Output

[illegible]

Viva Questions

1. What is scan conversion?
2. Why is the Midpoint Circle Algorithm preferred over the basic circle equation?
3. What is the decision parameter?
4. Explain eight-way symmetry.
5. Why is only one octant calculated?
6. Why are floating-point calculations avoided?
7. What is aliasing?
8. Compare Bresenham Circle and Midpoint Circle algorithms.
9. Why is the initial decision parameter equal to $1-r_1-r_1-r_1$?
10. What is raster graphics?
11. Can this algorithm draw ellipses?
12. Explain the role of OpenGL in this experiment.
13. What is the computational complexity?
14. Define graphics primitives.
15. Give industrial applications of circle drawing algorithms.

Assignment Questions

1. Modify the program to accept user-defined center coordinates.

2. Draw concentric circles.
3. Implement the Midpoint Ellipse Algorithm.
4. Animate a growing circle.
5. Draw Olympic rings using the algorithm.
6. Compare Midpoint Circle with the Polar Circle Algorithm.
7. Implement anti-aliased circle rendering.
8. Measure execution time for circles of varying radii.

EXPERIMENT – 3

IMPLEMENTATION OF TWO-DIMENSIONAL TRANSFORMATIONS USING HOMOGENEOUS COORDINATES

Aim

To implement the fundamental two-dimensional geometric transformations using homogeneous coordinate representation and transformation matrices in OpenGL.

Course Outcome

CO2: Apply two-dimensional graphics transformations using matrix representation and homogeneous coordinates for graphical object manipulation.

Bloom's Taxonomy Level

L3 – Apply

Learning Objectives

After completing this experiment, the student will be able to

- Understand coordinate transformations.
- Apply matrix operations in computer graphics.
- Implement translation, rotation, scaling, reflection and shearing.
- Understand homogeneous coordinates.
- Develop OpenGL programs for geometric transformations.

Output

Viva Questions

1. What is a geometric transformation?
2. Why are homogeneous coordinates used?
3. Explain the translation matrix.
4. What is the difference between scaling and shearing?
5. What is the purpose of reflection?
6. Why is rotation about the origin?
7. How can rotation about an arbitrary point be achieved?
8. What is the advantage of matrix representation?
9. What is the order of matrix multiplication in transformations?
10. Why is translation not represented by a 2×2 matrix?
11. What is the determinant of a rotation matrix?
12. Explain composite transformations.
13. What is affine transformation?
14. Give real-life applications of 2D transformations.
15. Compare 2D and 3D transformations.

Assignment Questions

1. Rotate a polygon about an arbitrary pivot point.
 2. Perform non-uniform scaling.
 3. Implement reflection about the line $y=x$.
 4. Create an interactive menu-driven transformation program.
 5. Animate continuous rotation of an object.
 6. Compare transformation matrices for 2D and 3D graphics.
 7. Implement all transformations using a single matrix multiplication function.
 8. Apply multiple transformations to a complex polygon.
-

EXPERIMENT – 4

IMPLEMENT COMPOSITE TRANSFORMATIONS USING MATRIX REPRESENTATION.

Aim

To implement the fundamental two-dimensional geometric transformations using homogeneous coordinate representation and transformation matrices in OpenGL.

Course Outcome

CO2: Apply two-dimensional graphics transformations using matrix representation and homogeneous coordinates for graphical object manipulation.

Bloom's Taxonomy Level

L3 – Apply

Learning Objectives

After completing this experiment, the student will be able to

- Understand coordinate transformations.
- Apply matrix operations in computer graphics.
- Implement translation, rotation, scaling, reflection and shearing.
- Understand homogeneous coordinates.
- Develop OpenGL programs for geometric transformations.

Output

Viva Questions

16. What is a geometric transformation?
17. Why are homogeneous coordinates used?
18. Explain the translation matrix.
19. What is the difference between scaling and shearing?
20. What is the purpose of reflection?
21. Why is rotation about the origin?
22. How can rotation about an arbitrary point be achieved?
23. What is the advantage of matrix representation?
24. What is the order of matrix multiplication in transformations?
25. Why is translation not represented by a 2×2 matrix?
26. What is the determinant of a rotation matrix?
27. Explain composite transformations.
28. What is affine transformation?
29. Give real-life applications of 2D transformations.
30. Compare 2D and 3D transformations.

Assignment Questions

9. Rotate a polygon about an arbitrary pivot point.
 10. Perform non-uniform scaling.
 11. Implement reflection about the line $y=x$.
 12. Create an interactive menu-driven transformation program.
 13. Animate continuous rotation of an object.
 14. Compare transformation matrices for 2D and 3D graphics.
 15. Implement all transformations using a single matrix multiplication function.
 16. Apply multiple transformations to a complex polygon.
-

EXPERIMENT – 5

IMPLEMENT THE COHEN–SUTHERLAND LINE CLIPPING ALGORITHM.

Aim

To implement the fundamental two-dimensional geometric transformations using homogeneous coordinate representation and transformation matrices in OpenGL.

Course Outcome

CO2: Apply two-dimensional graphics transformations using matrix representation and homogeneous coordinates for graphical object manipulation.

Bloom's Taxonomy Level

L3 – Apply

Learning Objectives

After completing this experiment, the student will be able to

- Understand coordinate transformations.
- Apply matrix operations in computer graphics.
- Implement translation, rotation, scaling, reflection and shearing.
- Understand homogeneous coordinates.
- Develop OpenGL programs for geometric transformations.

Output

Viva Questions

31. What is a geometric transformation?
32. Why are homogeneous coordinates used?

33. Explain the translation matrix.
34. What is the difference between scaling and shearing?
35. What is the purpose of reflection?
36. Why is rotation about the origin?
37. How can rotation about an arbitrary point be achieved?
38. What is the advantage of matrix representation?
39. What is the order of matrix multiplication in transformations?
40. Why is translation not represented by a 2×2 matrix?
41. What is the determinant of a rotation matrix?
42. Explain composite transformations.
43. What is affine transformation?
44. Give real-life applications of 2D transformations.
45. Compare 2D and 3D transformations.

Assignment Questions

17. Rotate a polygon about an arbitrary pivot point.
 18. Perform non-uniform scaling.
 19. Implement reflection about the line $y=x$.
 20. Create an interactive menu-driven transformation program.
 21. Animate continuous rotation of an object.
 22. Compare transformation matrices for 2D and 3D graphics.
 23. Implement all transformations using a single matrix multiplication function.
 24. Apply multiple transformations to a complex polygon.
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EXPERIMENT – 6

IMPLEMENT THE Z-BUFFER ALGORITHM FOR HIDDEN SURFACE REMOVAL.

Aim

To implement the fundamental two-dimensional geometric transformations using homogeneous coordinate representation and transformation matrices in OpenGL.

Course Outcome

CO2: Apply two-dimensional graphics transformations using matrix representation and homogeneous coordinates for graphical object manipulation.

Bloom's Taxonomy Level

L3 – Apply

Learning Objectives

After completing this experiment, the student will be able to

- Understand coordinate transformations.
- Apply matrix operations in computer graphics.
- Implement translation, rotation, scaling, reflection and shearing.
- Understand homogeneous coordinates.
- Develop OpenGL programs for geometric transformations.

Output

Viva Questions

46. What is a geometric transformation?
47. Why are homogeneous coordinates used?

48. Explain the translation matrix.
49. What is the difference between scaling and shearing?
50. What is the purpose of reflection?
51. Why is rotation about the origin?
52. How can rotation about an arbitrary point be achieved?
53. What is the advantage of matrix representation?
54. What is the order of matrix multiplication in transformations?
55. Why is translation not represented by a 2×2 matrix?
56. What is the determinant of a rotation matrix?
57. Explain composite transformations.
58. What is affine transformation?
59. Give real-life applications of 2D transformations.
60. Compare 2D and 3D transformations.

Assignment Questions

25. Rotate a polygon about an arbitrary pivot point.
 26. Perform non-uniform scaling.
 27. Implement reflection about the line $y=x$.
 28. Create an interactive menu-driven transformation program.
 29. Animate continuous rotation of an object.
 30. Compare transformation matrices for 2D and 3D graphics.
 31. Implement all transformations using a single matrix multiplication function.
 32. Apply multiple transformations to a complex polygon.
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EXPERIMENT – 7

IMPLEMENT THE PHONG ILLUMINATION MODEL FOR A 3D OBJECT.

Aim

To implement the fundamental two-dimensional geometric transformations using homogeneous coordinate representation and transformation matrices in OpenGL.

Course Outcome

CO2: Apply two-dimensional graphics transformations using matrix representation and homogeneous coordinates for graphical object manipulation.

Bloom's Taxonomy Level

L3 – Apply

Learning Objectives

After completing this experiment, the student will be able to

- Understand coordinate transformations.
- Apply matrix operations in computer graphics.
- Implement translation, rotation, scaling, reflection and shearing.
- Understand homogeneous coordinates.
- Develop OpenGL programs for geometric transformations.

Output

Viva Questions

61. What is a geometric transformation?
62. Why are homogeneous coordinates used?

63. Explain the translation matrix.
64. What is the difference between scaling and shearing?
65. What is the purpose of reflection?
66. Why is rotation about the origin?
67. How can rotation about an arbitrary point be achieved?
68. What is the advantage of matrix representation?
69. What is the order of matrix multiplication in transformations?
70. Why is translation not represented by a 2×2 matrix?
71. What is the determinant of a rotation matrix?
72. Explain composite transformations.
73. What is affine transformation?
74. Give real-life applications of 2D transformations.
75. Compare 2D and 3D transformations.

Assignment Questions

33. Rotate a polygon about an arbitrary pivot point.
 34. Perform non-uniform scaling.
 35. Implement reflection about the line $y=x$.
 36. Create an interactive menu-driven transformation program.
 37. Animate continuous rotation of an object.
 38. Compare transformation matrices for 2D and 3D graphics.
 39. Implement all transformations using a single matrix multiplication function.
 40. Apply multiple transformations to a complex polygon.
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EXPERIMENT – 8

DEVELOP A BASIC OPENGL PROGRAM USING VERTEX AND FRAGMENT SHADERS.

Aim

To implement the fundamental two-dimensional geometric transformations using homogeneous coordinate representation and transformation matrices in OpenGL.

Course Outcome

CO2: Apply two-dimensional graphics transformations using matrix representation and homogeneous coordinates for graphical object manipulation.

Bloom's Taxonomy Level

L3 – Apply

Learning Objectives

After completing this experiment, the student will be able to

- Understand coordinate transformations.
- Apply matrix operations in computer graphics.
- Implement translation, rotation, scaling, reflection and shearing.
- Understand homogeneous coordinates.
- Develop OpenGL programs for geometric transformations.

Output

Viva Questions

- 76. What is a geometric transformation?
- 77. Why are homogeneous coordinates used?

78. Explain the translation matrix.
79. What is the difference between scaling and shearing?
80. What is the purpose of reflection?
81. Why is rotation about the origin?
82. How can rotation about an arbitrary point be achieved?
83. What is the advantage of matrix representation?
84. What is the order of matrix multiplication in transformations?
85. Why is translation not represented by a 2×2 matrix?
86. What is the determinant of a rotation matrix?
87. Explain composite transformations.
88. What is affine transformation?
89. Give real-life applications of 2D transformations.
90. Compare 2D and 3D transformations.

Assignment Questions

41. Rotate a polygon about an arbitrary pivot point.
 42. Perform non-uniform scaling.
 43. Implement reflection about the line $y=x$.
 44. Create an interactive menu-driven transformation program.
 45. Animate continuous rotation of an object.
 46. Compare transformation matrices for 2D and 3D graphics.
 47. Implement all transformations using a single matrix multiplication function.
 48. Apply multiple transformations to a complex polygon.
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EXPERIMENT – 9

CREATE A SIMPLE INTERACTIVE 3D SCENE USING UNITY

Aim

To implement the fundamental two-dimensional geometric transformations using homogeneous coordinate representation and transformation matrices in OpenGL.

Course Outcome

CO2: Apply two-dimensional graphics transformations using matrix representation and homogeneous coordinates for graphical object manipulation.

Bloom's Taxonomy Level

L3 – Apply

Learning Objectives

After completing this experiment, the student will be able to

- Understand coordinate transformations.
- Apply matrix operations in computer graphics.
- Implement translation, rotation, scaling, reflection and shearing.
- Understand homogeneous coordinates.
- Develop OpenGL programs for geometric transformations.

Output

Viva Questions

91. What is a geometric transformation?
92. Why are homogeneous coordinates used?

93. Explain the translation matrix.
94. What is the difference between scaling and shearing?
95. What is the purpose of reflection?
96. Why is rotation about the origin?
97. How can rotation about an arbitrary point be achieved?
98. What is the advantage of matrix representation?
99. What is the order of matrix multiplication in transformations?
100. Why is translation not represented by a 2×2 matrix?
101. What is the determinant of a rotation matrix?
102. Explain composite transformations.
103. What is affine transformation?
104. Give real-life applications of 2D transformations.
105. Compare 2D and 3D transformations.

Assignment Questions

49. Rotate a polygon about an arbitrary pivot point.
 50. Perform non-uniform scaling.
 51. Implement reflection about the line $y=x$.
 52. Create an interactive menu-driven transformation program.
 53. Animate continuous rotation of an object.
 54. Compare transformation matrices for 2D and 3D graphics.
 55. Implement all transformations using a single matrix multiplication function.
 56. Apply multiple transformations to a complex polygon.
-

EXPERIMENT – 10

TO IMPLEMENT THE 3D VIEWING PIPELINE, INCLUDING PROJECTION AND VIEWPORT MAPPING FOR DISPLAYING 3D OBJECTS ON A 2D SCREEN.

Aim

To implement the fundamental two-dimensional geometric transformations using homogeneous coordinate representation and transformation matrices in OpenGL.

Course Outcome

CO2: Apply two-dimensional graphics transformations using matrix representation and homogeneous coordinates for graphical object manipulation.

Bloom's Taxonomy Level

L3 – Apply

Learning Objectives

After completing this experiment, the student will be able to

- Understand coordinate transformations.
 - Apply matrix operations in computer graphics.
 - Implement translation, rotation, scaling, reflection and shearing.
 - Understand homogeneous coordinates.
 - Develop OpenGL programs for geometric transformations.
-

Output

Viva Questions

106. What is a geometric transformation?
107. Why are homogeneous coordinates used?
108. Explain the translation matrix.
109. What is the difference between scaling and shearing?
110. What is the purpose of reflection?
111. Why is rotation about the origin?
112. How can rotation about an arbitrary point be achieved?
113. What is the advantage of matrix representation?
114. What is the order of matrix multiplication in transformations?
115. Why is translation not represented by a 2×2 matrix?
116. What is the determinant of a rotation matrix?
117. Explain composite transformations.
118. What is affine transformation?
119. Give real-life applications of 2D transformations.
120. Compare 2D and 3D transformations.

Assignment Questions

57. Rotate a polygon about an arbitrary pivot point.
 58. Perform non-uniform scaling.
 59. Implement reflection about the line $y=x$.
 60. Create an interactive menu-driven transformation program.
 61. Animate continuous rotation of an object.
 62. Compare transformation matrices for 2D and 3D graphics.
 63. Implement all transformations using a single matrix multiplication function.
 64. Apply multiple transformations to a complex polygon.
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